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Euclides would be proud: setting up basic geometry to project rays in your galaxy!

1. GEOMETRY SETUP OF GALAXY HALOS

We are using TRIDENT software to compute rays that go through each simulated halo along a distance that is twice the virial radius of the galaxy, which is calculated using

$$\Delta_c \rho_c \left(\frac{4}{3} \pi R_{vir}^3 \right) = M_{halo}, \tag{1}$$

where ρ_c is the critical density of the universe (at the respective redshift) and Δ_c is a factor derived from the relaxation of dark matter halos after spherical collapse, which depends on the cosmology. In what follows, we assume Bryan & Norman (1998) fit for Δ_c for $\Omega_{Rad} = 0$.

A new coordinate basis (different from the simulations coordinates) is defined to trace the rays, and it is depicted in Fig. 1. We start by calculating the angular momentum of the stars component of each subhalo (or galaxy) and then normalize it. We define

$$\vec{L_N} = \vec{L} \times \hat{z}, \tag{2}$$

where $\hat{z}$ is the vector $(0, 0, 1)$, defined in the simulations coordinates system. If we want rays to go through the halo at a certain α angle of inclination, rays need to be parallel to the following **projection vector ($\vec{P}$)**

$$\vec{P} = \sin(\alpha)\vec{L} + \cos(\alpha)\vec{L_N}. \tag{3}$$

For visualization purposes, we compute the **north vector ($\vec{N}$)** as

$$\vec{N} = \cos(\alpha)\vec{L} - \sin(\alpha)\vec{L_N}, \tag{4}$$

this vector is defined such that for an edge-on simulation ($\alpha = 0$), angular momentum points upwards (the y-direction of an EW map for example). Finally, a **west vector** is simply defined as $\vec{W} = \vec{P} \times \vec{N}$. These three vectors define a new coordinate basis $\vec{P} - \vec{N} - \vec{W}$ in which rays are created. In this new coordinate basis, rays are parallel to the $\vec{P}$ vector and orthogonal to the plane defined by the vectors $\vec{N} - \vec{W}$.

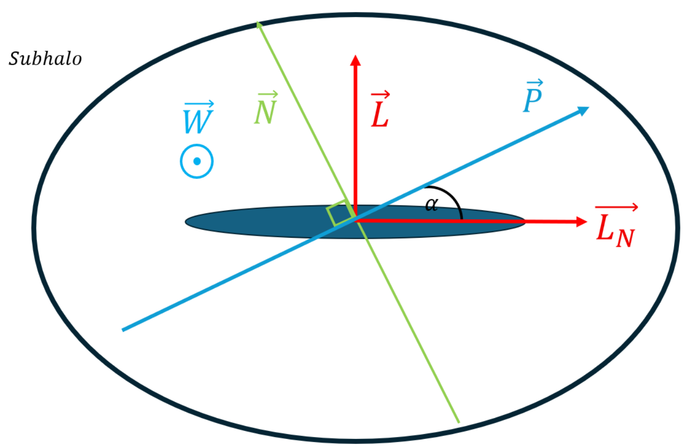

Figure 1. Depiction of the geometry employed to define the *rays-plane* as explained in the text. Blue ellipse illustrates the galaxy in the center and black external contour represents the halo extension up to the virial radius. Rays are constructed parallel to vector $\vec{P}$.